Lab 2

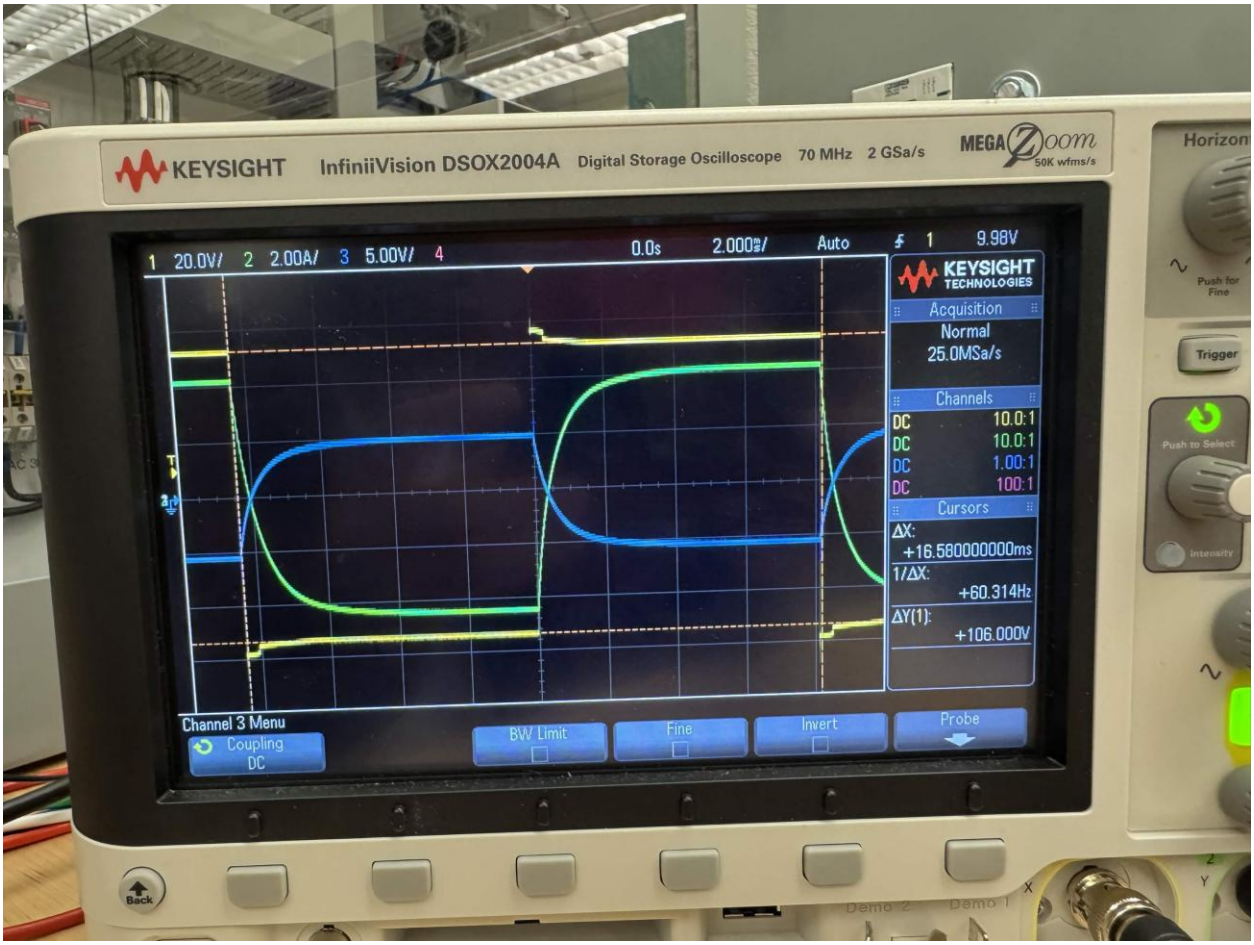
Section B01
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Experiment 1

Explain the shape of i_L using the load time constant L/R, and compare it with your pre-lab sketch



$$\tau = \frac{L}{R} = \frac{5mH}{12.5\Omega} = 0.4ms$$

The time constant is the time it takes for the inductor current to charge up to 63% of its maximum current output. We can see this in the waveform, with the current charging in the positive and negative direction based on the voltage within the period interval. Within $5\tau = 2ms$, the graph reaches 99% of the maximum. In comparison to the pre-lab sketch, the experimental graph is accurate in terms of the timing.

Record, $I_{L,1}$, $I_{L,3}$, $I_{L,5}$, $I_{L,7}$, the THD if i_L and the power delivered to R. Compare $I_{L,1}$, P_1 and $I_{L,3}$ with your pre-lab values and account for any deviation

R = 12.5

VRMS = 47.8V at f=60Hz

α = 0

Comparison Table

Quantity	**Pre-lab**	**Measured**	**Deviation**
IL,1	3.917 A	3.95 A	0.8 %
IL,3	1.203 A	1.03 A	−13.7 %
IL,5	N/A	0.51 A	N/A
IL,7	N/A	0.3 A	N/A
THD of iL	37.5 %	31.1 %	N/A
P1 = (IL,1^2)R	191.9 W	195.031 W	1.68%
Total power in R	218.8 W	188.8 W	N/A

Deviation/Comparison:

- IL1:
  - Predicted = 3.917A,
  - Measured = 3.95A
  - Deviation = (3.95-3.917)/3.917 * 100 = 0.8**%**
- IL3:
  - Predicted = 1.203A
  - Measured = 1.03A
  - Deviation = (1.03-1.203)/1.203 *100 = -**13.7%**
- P1:
  - Predicted = 194.8 W
  - Measured = 195.031 W
  - Deviation = (195.031-191.8)/191.8 * 100 = 1.68**%**
- We notice here that both currents come in below prediction, this is because the IGBTS, diodes and other components are not ideal, there will be leakage that causes voltage drop and the vt swings will not be a perfect +-55V. This difference in voltage can cause the fundamental and harmonic amplitudes to shrink which corresponds with the lower than predicted pattern seen above.

- We can see that power also comes in lower too as it is proportional to current, the formula shows that P is proportional to IL1^2, so the reasoning of unideal components matches.

Comparing our prelab calculations and our experimental results, our values match closely with each other. Our experimental values are slightly greater except IL,3, which can be due to any conduction loss in the switch, and the increase in value due to distortion voltages and currents affecting the output.

Experiment 2

At your pre-lab : record $I_{L,1}$, $I_{L,3}$, $I_{L,5}$, $I_{L,7}$, the THD of i_L and the total power in R.

Lab Measurements:

α	IL,1	IL,3	IL,5	IL,7	THD	Total Power in R
0	3.95A	1.03A	0.51A	0.3A	31.31%	195.031W
0.257 (Prelab value)	2.56A	0.77A	0.28A	0.26A	32.6%	81.92W

α = 0.167

- Vrms = 32.6V

α = 0.257

- Vrms = 39.5V

Compare $I_{L,1}$, P_1 and $I_{L,3}$ with question 2, and compare the total power in R with P_1.

Comparing IL,1 IL,3, P1 at **α = 0.257**

	Pre-Lab Calculation	Experimental
IL,1	2.708A	2.56A
IL,3	0.905A	0.77A
P1	91.97W	81.92W

- Looking at these numbers, our values are consistently below the pre-lab calculations: IL,1 is low by 5.5%, P1 by 10.6%, and IL,3 by 14.9%. This is expected, since the pre-lab calculation assumes an ideal bridge that switches instantaneously, while the real H-bridge inserts a brief dead time at every switching transition (both devices in the leg off, to prevent shoot-through). This clips a small number of volt-seconds out of each pulse, so the actual voltage delivered to the load comes in slightly below the ideal phase-shift prediction. IL,3 deviates more than IL,1 because that same fixed dead-time loss is a larger relative cut into the smaller 3rd-harmonic voltage than into the fundamental. P1's deviation (10.6%) is also consistent with IL,1's, since P1 proportional $IL,1^2$ means a 5.5% current error should roughly double to about 10.7% in power, which matches closely.

At α = 1/6: record the same quantities. Compare the 3rd harmonic current with Experiment 1 and explain the result, including anything that remains. Tabulate $I_{L,1}$, $I_{L,3}$, $I_{L,5}$, $I_{L,7}$ and the THD of i_L at α = 0, 1/6 and your pre-lab value, and use these harmonic currents to explain how the THD varies with α.

α	IL,1	IL,3	IL,5	IL,7	THD	Total Power in R
0.167	3.24A	0.04A	0.42A	0.26A	16.0%	129W

Experiment 1 Values for comparison:

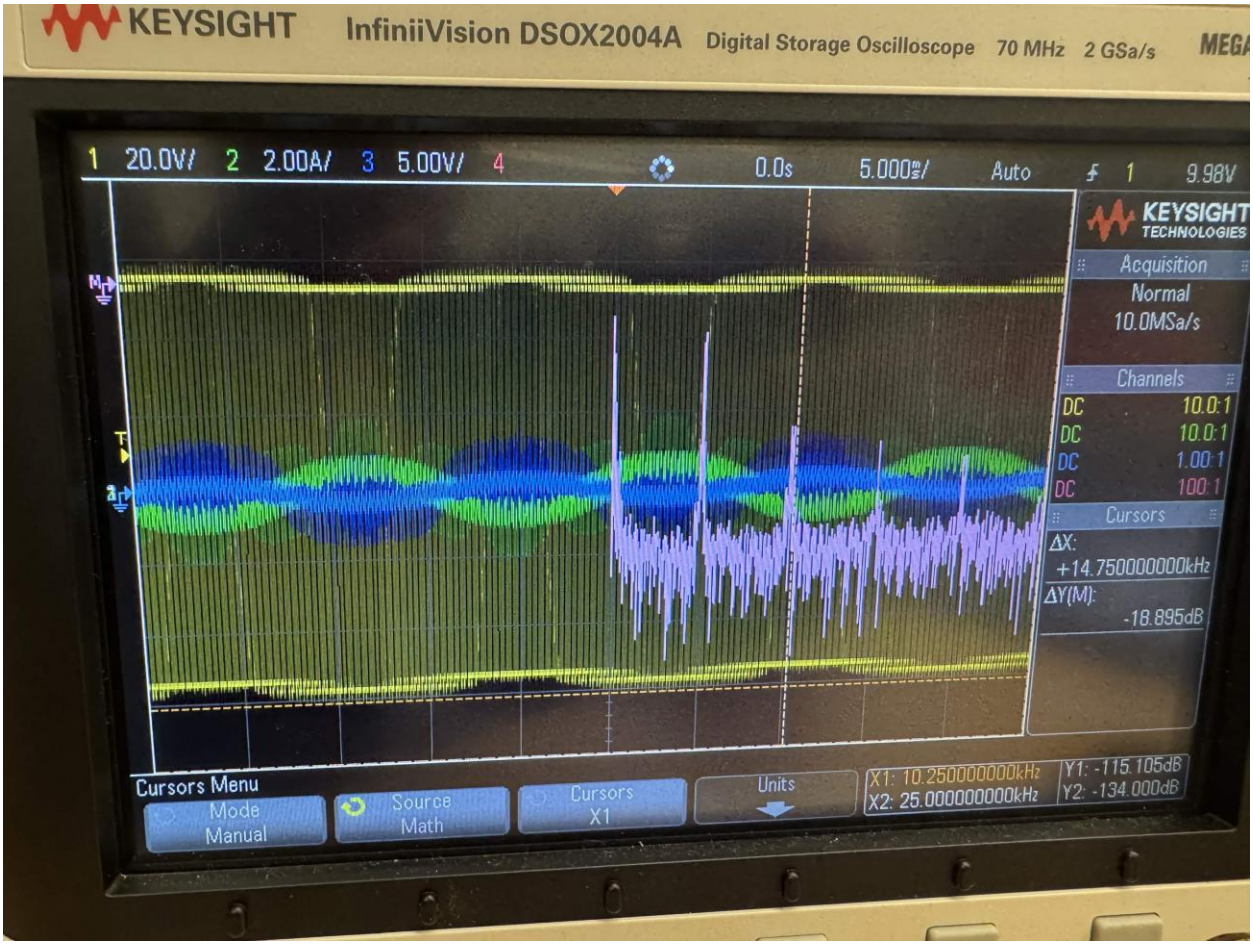
IL,3 :

- Measured: 1.03A
- Prelab: 1.203A

Comparing the third harmonic between Experiment 1 and α=1/6, IL,3 drops from 1.03A to 0.04A. This matches the phase shift harmonic formula, $\hat{V}t,3 = (4vd/3\pi)\cdot|\cos(3\pi\alpha)|$: at α=1/6, $\cos(3\pi/6) = \cos(\pi/2) = 0$ exactly, so the theoretical prediction is a complete null of the 3rd harmonic voltage. That's why we see (almost) zero current at the 3rd harmonic here: IL,3 = ˆ Vt / Z3, so if there's no 3rd-harmonic voltage being produced by the bridge, there's nothing to drive a 3rd-harmonic current through the load. The 0.04A that remains comes from two practical limits the ideal formula doesn't account for: the firmware only applies α in steps of 1/120, so the actual α used is slightly off from exactly 1/6, and the 5μs dead time at every switching transition clips a small amount of volt-seconds that the ideal formula assumes doesn't exist. Both push cos(3πα) away from exactly zero, leaving a small residual current rather than a perfect null.

Experiment 3

Compare the waveforms of V_t and V_R and explain the difference



V_t is the yellow (20V/div) and V_R is the blue (5V/div). V_t swings from +55 and -55V, as it is the carrier signal. V_R is the reference signal closely follows our desired amplitude and frequency.

Using the oscilloscope FFT of i_L, measure the frequency and rms value of the three largest components near f_{sw} and the two largest near $2f_{sw}$, and compare the largest with question 3(a).

Largest components near f_{sw} (m_i = 0.2)

Component	Frequency	Current (Rms)
1	10.15kHz	25mA
2	14.97kHz	28mA
3	19.93kHz	10mA

Largest components near $2f_{sw}$(For m_i = 0.2)

Component	Frequency	Current (Rms)
1	9.75khz	23mA
2	10.25kHz	23mA

Largest components near f_{sw} (For m_i = 0.88)

Component	Frequency	Current (Rms)
1	10.15kHz	33mA
2	14.97kHz	13.5mA
3	19.93kHz	6.7mA

Largest components near $2f_{sw}$(For m_i = 0.88)

Component	Frequency	Current (Rms)
1	9.75khz	33mA
2	10.25kHz	33mA

Using your oscilloscope FFT readings, estimate the THD of i_L including the components near f_{sw}, and compare it with the THD the Fluke 39 reports. Explain the difference.

For m_i = 0.2

THD = $\sqrt{[(0.025)^2 + (0.028)^2 + (0.01)^2]} / 0.62 * 100\%$

THD = 6.27%

Fluke Reading: THD = 13.2%

For m_i = 0.88

THD = $\sqrt{[(0.033)^2 + (0.0135)^2 + (0.0067)^2]} / 2.71 * 100\%$

THD = 1.34%

Fluke Reading: THD = 3.1%

The oscilloscope can measure components around f_{sw} and higher harmonics while the Fluke39 cannot, hence the THD value being different from one another. The difference is largest at m_i = 0.2 because the switching ripple stays roughly the same size while the fundamental is about five time smaller.

Compare the low-order harmonic currents on the Fluke 39 (up to the 31st) with what ideal SPWM predicts. If there are components the theory does not predict, propose at least two possible causes in the circuit or the modulator. After the lab, test the cause you consider most likely by changing SPWM_bipolar.plecs, and report the result.

Low Order Harmonic Fluke Readings (for mi = 0.2):

$I_{L,1}$ = 0.43A

$I_{L,3}$ = 0.05A

$I_{L,5}$ = 0.02A

$I_{L,7}$ = 0.01A

Low Order Harmonic Fluke Readings (for mi = 0.2):

$I_{L,1}$ = 2.34A

$I_{L,3}$ = 0.06A

$I_{L,5}$ = not recorded (we think close 0)

$I_{L,7}$ = not recorded (we think close 0)

Two possible causes:
- During each 5μs blanking interval both devices in a leg are off, and the leg voltage is set by whichever diode carries i_L. Each leg loses $t_d \cdot f_{sw} \cdot V_D$ = 5μs · 4.98kHz · 55 V = 1.37 V on average, opposing the current, so V_t carries a 2.7V square-wave error in phase i_L. This lowers the fundamental of V_t by (4/π) · 2.7 = 3.5 V peak and adds odd low-order harmonics. In part A there were only two transitions per period, so it didn't matter there, but SPWM has 83.

- IGBT and diode voltage drops about 2V in total, which also forms a square wave in phase with i_L and has the same effect as the dead time.

Results from the simulation: Dead time lowers $I_{L,1}$ by about 0.2 A at m_i = 0.88 and gives $I_{L,3}$ = 0.06A, $I_{L,5}$ = 0.03A and $I_{L,7}$ = 0.02A. Ideal switches don't model the device drops, so the simulated $I_{L,1}$ should stay a bit above the measured value.

Experiment 4

Tabulate P, $V_{t,rms}$, $I_{L,rms}$, the power factor $P/(V_{t,rms}I_{L,rms})$ and the load's displacement factor $\cos\Phi_1$ at the three operating points, and compare the $m_i = 0.88$, 12.5Ω point with question 3(b).

mi	R	P	Vt RMS	IL RMS	PF	$cos\phi$
0.88	12.5	-77.01W	54.278V	2.2165A	0.6401	0.606
0.44	12.5	-22.64W	54.631V	1.0283A	0.4030	0.177
0.88	50	-107.47W	53.782V	0.7153A	2.7936	0.854

Explain how the bridge-output power factor compares with $\cos\Phi_1$, and how it changes with m_i and with R.

Calculations for PF and $cos\phi$ at m_i = 0.88:

$$PF = \frac{P}{V_{t,rms}\, I_{t,rms}}$$

$$PF = \frac{|-77.01|}{(54.278)\,(2.2165)}$$

$$PF = 0.6401$$

$I_{L,1}$ = 2.34A

$$cos\phi = \frac{PF}{\left(\frac{I_1}{I_{rms}}\right)}$$

$$cos\phi = \frac{0.6401}{\left(\frac{2.34}{2.2165}\right)}$$

$$cos\phi = 0.606$$

PF ≈ $(V_{t,1}/V_{t,rms})(I_{L,1}/I_{L,rms})$ cos ϕ_1. The current is nearly sinusoidal and cos $\phi_1 \approx 0.99$, so the low PF comes from the voltage: in bipolar SPWM $V_{t,rms} = v_d$ at any m_i, while $V_{t,1} = m_i v_d/\sqrt{2}$.

The switching frequency voltage adds volt amperes but almost no power, since the inductor blocks its current. Hence PF ≈ $m_i/\sqrt{2}$, well below cos ϕ_1.

Halving m_i roughly halves the PF (0.529 to 0.333; ideal 0.617 to 0.315). Raising R to 50 Ω increases cos ϕ_1 to 0.999, but the ripple current (set by L) stays the same while $I_{L,1}$ falls by a factor of 4, so the current becomes more distorted. The PF therefore changes little (0.577; ideal 0.649).

Experiment 5

We ran out of time

Comparison

All tables are recorded at R = 12.5ohms

Square Wave (Experiment 1) Table:

$I_{L,1}$	P_1	Largest low-order $I_{L,x}$
3.95A	195.031W	$I_{L,3}$ = 1.03A

Phase Shift at Pre-lab value (Experiment 2) Table - $\alpha = 0.257$:

$I_{L,1}$	P_1	Largest low-order $I_{L,x}$
2.56	81.92W	$I_{L,3}$ = 0.77A

Bipolar SPWM at mi = 0.88 (Experiment 3) Table:

$I_{L,1}$	P_1	Largest low-order $I_{L,x}$	Largest component near f_{sw} or $2f_{sw}$
3.24A	131.22W	$I_{L,5}$= 0.42A	33mA

Unipolar SPWM Table -> Unavailable since we didn't have time to conduct experiment 5

Based on the experimental results, bipolar SPWM produced the lowest dominant low-order harmonic current relative to its fundamental, followed by the square-wave and phase-shift modulation. Square-wave modulation had a third-harmonic current of 1.03 A, while phase-shift modulation reduced it to 0.77 A but also reduced the fundamental current. Bipolar SPWM shifts most of its harmonic content to higher frequencies, where the inductor provides greater impedance and reduces harmonic currents. Unipolar SPWM is

theoretically expected to have even lower harmonic distortion since its dominant harmonics occur near twice the switching frequency, although this could not be verified because Experiment 5 was not completed. For square-wave and phase-shift modulation, each switch turns ON once per fundamental period, while bipolar and unipolar SPWM require approximately 83 turn-ON events per switch per period. Therefore, PWM methods generally provide better harmonic performance at the cost of increased switching losses, while unipolar SPWM offers improved harmonic reduction without increasing the individual switch frequency compared to bipolar SPWM.